

SGA 3, Exposé XIX
Reductive Groups—Generalities
English Loop-1 Reconstruction

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EXPOSÉ XIX

REDUCTIVE GROUPS—GENERALITIES

by *M. Demazure*

The remainder of this Seminar is devoted to the study of reductive groups. Its principal aim is to generalize Chevalley’s classical results (the *Bible* and *Tôhoku*)¹ to arbitrary base schemes. The two central results are the uniqueness theorem (Exposé XXIII, Theorem 4.1 and Corollaries 5.1–5.10) and the existence theorem (Exposé XXV, Theorem 1.1) for pinned reductive group schemes corresponding to prescribed root data. The proofs used here are inspired by Chevalley’s; the language of schemes makes it possible to give them greater effectiveness.

The results of the first volume of the *Bible* (Exposés 1–13) will be used systematically. By contrast, we shall prove directly over an arbitrary scheme the results of the second volume (in particular the isomorphism theorem); knowledge of the proofs over an algebraically closed field is therefore not absolutely indispensable.

In proving these two fundamental results, we shall use only the most elementary results from the theory of groups of multiplicative type, found essentially in Exposés VIII and IX; on the other hand, we shall make essential use of the results of Exposé XVIII.² The reader interested specifically in the existence and uniqueness theorems may, on a first reading, skip Exposés X–XVII.

1. Review of groups over an algebraically closed field

1.1. Throughout this subsection, k will denote an algebraically closed field. As announced above, the only results from the *Bible* used below are contained in its first volume (Exposés 1–13).

All the results of the *Bible* (*loc. cit.*) concerning semisimple groups hold more generally for reductive groups (defined below), and their proofs are identical after the following harmless modifications:

- Exposé 9, §4, Definition 3: see 1.6.1 below;

⁰Editor’s note: version of 13 October 2024.

¹Editor’s note: see the bibliographical references at the end of this Exposé. In particular, the 2005 re-edition of the Chevalley Seminar 1956–1958, cited as *Bible*, was revised by P. Cartier; see [Ch05].

²Editor’s note: more precisely, Proposition XVIII 2.3 (extension of a “generic homomorphism” between groups) is used in XXII 4.1.11 and then in the proof of uniqueness in XXIII, as well as in XXIV. Theorem XVIII 3.7 (construction of a group from a group germ) is used only in XXV.

- Exposé 12, §4, Theorem 2(e): omit “finite”;
- Exposé 13, §3, Theorem 2: replace “rank” by “semisimple rank”;
- Exposé 13, §4, Corollary 2 to Theorem 3: replace “rank” by “reductive rank”.

1.2. Let G be a smooth affine connected k -group. The *radical* of G (*Bible*, §9.4, Proposition 2)³ is the reduced subgroup associated with the identity component of the intersection of the Borel subgroups of G ; it is also the largest smooth connected solvable normal subgroup of G . We denote it by $\text{rad}(G)$.

The *unipotent radical* of G is the unipotent part of the radical of G ; it is also the largest smooth connected unipotent normal subgroup of G . We denote it by $\text{rad}^u(G)$.

1.3. Let G be a smooth affine connected k -group, and let Q be a torus in G . Then the centralizer $\text{Centr}_G(Q)$ of Q in G is a closed subgroup of G (Exposé VIII, 6.7), smooth (Exposé XI, 2.4), and connected (cf. *Bible*, §6.6, Theorem 6(a), or [Ch05], §6.7, Theorem 7(a)).

We have the fundamental relation

$$(1.3.1) \quad \text{rad}^u(\text{Centr}_G(Q)) = \text{rad}^u(G) \cap \text{Centr}_G(Q).$$

First,⁴

$$U = \text{rad}^u(G) \cap \text{Centr}_G(Q)$$

is a normal unipotent subgroup of $\text{Centr}_G(Q)$; let us show that it is smooth and connected. Let Q act on $\text{rad}^u(G)$ by inner automorphisms. Then U is precisely the scheme of fixed points of this action. This scheme is smooth and connected by Lemma 1.4 below.

Consequently, U is a closed subgroup of $\text{rad}^u(\text{Centr}_G(Q))$. On the other hand, by the *Bible*, Exposé 12, §3, Corollary to Theorem 1,

$$\text{rad}^u(\text{Centr}_G(Q))(k) \subset \text{rad}^u(G)(k).$$

Equality (1.3.1) follows.

Let us record one special case of the preceding result. If G is a smooth affine connected k -group and T is a maximal torus of G , then

$$(1.3.2) \quad \text{Centr}_G(T) = T \cdot (\text{Centr}_G(T) \cap \text{rad}^u(G)).$$

Indeed (cf. *Bible*, §6.6, Theorem 6(c), or [Ch05], §6.7, Theorem 7(c)), $\text{Centr}_G(T)$ is solvable and hence is the semidirect product of its maximal torus T and its unipotent radical.

By the density theorem (cf. *Bible*, §6.5, Theorem 5(a), or [Ch05], §6.6, Theorem 6(a)), the union of $\text{Centr}_G(T)$, as T ranges over the maximal tori of G , contains a dense open subset of G . It therefore follows from (1.3.2) that:

Corollary 1.3.3.— *If G is a smooth affine connected k -group, then the union of $T \cdot \text{rad}^u(G)$, where T ranges over the maximal tori of G , contains a dense open subset of G .*

Notation 1.4.0.—⁵ *We shall systematically use the following notation. If S is a scheme and s is a point of S , then $\bar{s}$ denotes the spectrum of an algebraic closure $\kappa(\bar{s})$ of $\kappa(s)$.*

Lemma 1.4.—⁵ *Let S be a scheme and let Q be an S -torus acting on an S -group scheme H , separated and smooth over S .*

³Editor’s note: the reference 9-06 (that is, Exposé 9, p. 6) has been replaced by §9.4 (that is, Exposé 9, §4), which applies both to the *Bible* and to [Ch05]. When the numbering in [Ch05] differs from that in the *Bible*, as happens in Exposé 6, both references will be given explicitly.

⁴Editor’s note: the argument that follows has been expanded.

⁵Editor’s note: this notation, which appeared in 2.3 of the original, has been inserted here; moreover, the statement and proof of 1.4 have been expanded.

- (i) The fixed-point functor H^Q is representable by a closed subscheme of H , smooth over S .
- (ii) If, moreover, H is affine over S and has connected fibers, then the same holds for H^Q .

Proof. First, by VIII 6.5(d),⁶ since H is separated over S and Q is essentially free over S , H^Q is representable by a closed subscheme of H . In particular, if H is affine over S , then so is H^Q .

Now consider the semidirect product $G = H \cdot Q$; it is smooth and separated over S . The centralizer $\underline{\text{Centr}}_G(Q)$ is then representable by a closed subscheme of G , of finite presentation over G , by Exposé XI 6.11(a),⁶ and it is smooth over S by Exposé XI 2.4.

Since

$$\underline{\text{Centr}}_G(Q) = H^Q \cdot Q,$$

it follows that H^Q is smooth over S . Indeed, using the section

$$\sigma : H^Q \longrightarrow \underline{\text{Centr}}_G(Q)$$

deduced from the identity section ϵ_Q of Q , one sees at once that H^Q is formally smooth over S ; and since ϵ_Q is of finite presentation, H^Q is locally of finite presentation over S . This proves (i).

Finally, suppose that H is affine over S and has connected fibers. Then every geometric fiber $G_{\bar{s}}$ of G is a smooth affine connected $\kappa(\bar{s})$ -group. Hence, by the first assertion of 1.3, the group

$$\underline{\text{Centr}}_{G_{\bar{s}}}(Q_{\bar{s}}) = (\underline{\text{Centr}}_G(Q))_{\bar{s}} = (H^Q)_{\bar{s}} \cdot Q_{\bar{s}}$$

is also connected, and this implies that $(H^Q)_{\bar{s}}$ is connected. $\square$

1.5. Recall that the *reductive rank* of a smooth affine k -group G is the common dimension of its maximal tori. We denote it by $\text{rgred}(G/k)$ or $\text{rgred}(G)$. The equality $\text{rgred}(G/k) = 0$ holds if and only if G is unipotent (that is, $G = \text{rad}^u(G)$), by the *Bible*, §6.4, Corollary 1 to Theorem 4, or [Ch05], §6.5, Corollary 1 to Theorem 5.

If H is a normal subgroup of the smooth affine k -group G , then the quotient G/H is affine and smooth (Exposé VI_B, 11.17 and VI_A, 3.3.2(iii)). Moreover (*Bible*, §7.3, Theorem 3(a),(c)),

$$\text{rgred}(G) = \text{rgred}(G/H) + \text{rgred}(H).$$

Definition 1.6.1.⁷ A k -group G is said to be *reductive* if it is smooth, affine, and connected, and if $\text{rad}(G)$ is a torus; that is, if

$$\text{rad}^u(G) = \{e\}.$$

Lemma 1.6.2.— Let G be a reductive k -group.

- (i) If Q is a torus of G , then $\underline{\text{Centr}}_G(Q)$ is reductive.
- (ii) In particular, if T is a maximal torus of G , then $\underline{\text{Centr}}_G(T) = T$.

⁶Editor's note: see also the additions made in VI_B, 6.2.4(d) and 6.5.5(a).

⁷Editor's note: the former number 1.6 has been replaced by numbers 1.6.1 through 1.6.3. Note that in this Seminar every reductive k -group (respectively, every semisimple k -group; cf. 1.8) is connected by definition.

- (iii) The center of G is contained in every maximal torus and is therefore diagonalizable.
- (iv) The radical of G is the unique maximal torus of $\underline{\text{Centr}}(G)$.

Proof. Indeed, (i) follows from (1.3.1), (ii) from (1.3.2), and (iii) follows from (ii). Finally, the maximal torus of $\underline{\text{Centr}}(G)$, that is, the neutral component $\underline{\text{Centr}}(G)^0$, is a smooth connected solvable subgroup normal in G , and hence is contained in $\text{rad}(G)$. Conversely, $\text{rad}(G)$ is a torus normal in G , and hence central (*Bible*, §4.3, Corollary to Proposition 2); this proves (iv). $\square$

Remark 1.6.3.— If G is reductive and $\dim(G) \neq \text{rgred}(G)$, then

$$\dim(G) - \text{rgred}(G) \geq 2.$$

Indeed, this difference is always even (cf. 1.10 below).

1.7. Let G be a smooth affine connected k -group, and let H be a smooth connected normal subgroup. Then

$$(1.7.1) \quad \text{rad}(H) = (\text{rad}(G) \cap H)_{\text{red}}^0, \quad \text{rad}^u(H) = (\text{rad}^u(G) \cap H)_{\text{red}}^0,$$

as is seen immediately. In particular, if G is reductive, then so is H .

Let $f : G \rightarrow G'$ be a surjective morphism of smooth affine connected k -groups. Then

$$(1.7.2) \quad f(\text{rad}^u(G)) = \text{rad}^u(G').$$

In particular, if G is reductive, then so is G' .

Let us prove (1.7.2). First, $f(\text{rad}^u(G))$ is contained in $\text{rad}^u(G')$. On introducing

$$H = (f^{-1}(\text{rad}^u(G'))_{\text{red}})^0,$$

which contains $\text{rad}^u(G)$, we have $\text{rad}^u(H) = \text{rad}^u(G)$; thus we are reduced to the case $G = H$, that is, to the case in which G' is unipotent. As T runs over the maximal tori of G , the union of the groups $T \text{rad}^u(G)$ is dense in G . Hence the union of the groups $f(T) f(\text{rad}^u(G))$ is dense in G' . But $f(T)$ consists of semisimple elements, so $f(T) = \{e\}$, since G' is unipotent. It follows that $f(\text{rad}^u(G))$ is dense in G' . Therefore, by *Bible*, §5.4, Lemma 4, or [Ch05], §6.1, Lemma 1,⁸ $f(\text{rad}^u(G))$ is an open subgroup of G' . Since G' is connected, it follows that

$$f(\text{rad}^u(G)) = G'.$$

N.B. Under the same hypotheses one can prove that

$$f(\text{rad}(G)) = \text{rad}(G').$$

1.8. A k -group G is said to be *semisimple* if it is smooth, affine, and connected, and if

$$\text{rad}(G) = \{e\},$$

that is, if G is reductive and $\underline{\text{Centr}}(G)$ is finite. If G is an arbitrary smooth affine connected k -group, then $G/\text{rad}(G)$ is semisimple (*Bible*, §9.4, Proposition 2), and $G/\text{rad}^u(G)$ is reductive. The *semisimple rank* of G , denoted by $\text{rgss}(G/k)$, or simply by $\text{rgss}(G)$, is the reductive rank of $G/\text{rad}(G)$.

⁸Editor's note: the following argument has been supplied in detail.

Thus, if G is reductive,

$$\mathrm{rgred}(G) = \mathrm{rgss}(G) + \dim(\mathrm{rad}(G)).$$

If G is a smooth affine connected k -group and Q is a central subtorus of G , then G/Q is semisimple if and only if G is reductive and $Q = \mathrm{rad}(G)$. Indeed, if G/Q is semisimple, then $Q \supset \mathrm{rad}(G)$; hence $\mathrm{rad}(G)$ is a torus, so G is reductive, and $\mathrm{rad}(G)$ is the maximal torus of $\underline{\mathrm{Centr}}(G)$, whence $\mathrm{rad}(G) = Q$. If G is reductive and Q is a central subgroup, then G/Q is reductive and

$$\mathrm{rgss}(G) = \mathrm{rgss}(G/Q).$$

1.9. Let K be an algebraically closed extension of k , and let G be a smooth affine connected k -group. Then G is reductive (respectively, semisimple) if and only if G_K is, and

$$\mathrm{rgred}(G/k) = \mathrm{rgred}(G_K/K), \quad \mathrm{rgss}(G/k) = \mathrm{rgss}(G_K/K).$$

1.10. Let G be a smooth connected k -group and let T be a torus of G .⁹ Write $\mathfrak{g}$ for the k -Lie algebra of G , that is, $\mathfrak{g} = \mathrm{Lie}(G) = \mathrm{Lie}(G/k)(k)$, and similarly write $\mathfrak{t} = \mathrm{Lie}(T)$. Under the action of T , through $T \rightarrow G$ and the adjoint representation of G , the Lie algebra $\mathfrak{g}$ decomposes as

$$(1.10.1) \quad \mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha,$$

where the elements $\alpha \in R$ are nontrivial characters of T and the spaces $\mathfrak{g}^\alpha$ are nonzero. The characters $\alpha \in R$ are called the *roots of G relative to T* . By Exposé II, 5.2.3(i),

$$(1.10.2) \quad \mathfrak{g}^0 = \mathrm{Lie}(\underline{\mathrm{Centr}}_G(T)).$$

In particular,¹⁰ since $\underline{\mathrm{Centr}}_G(T)$ is connected (by 1.3 when G is affine, and by Exposé XII, 6.6(b), in general), T is its own centralizer if and only if $\mathfrak{g}^0 = \mathrm{Lie}(T)$. This condition implies that T is maximal and that $\underline{\mathrm{Centr}}(G) \subset T$; hence, by Exposé XII, 8.8(d),

$$(1.10.3) \quad \underline{\mathrm{Centr}}(G) = \mathrm{Ker}(T \rightarrow \mathrm{GL}(\mathfrak{g})) = \bigcap_{\alpha \in R} \mathrm{Ker}(\alpha).$$

Moreover, $\underline{\mathrm{Centr}}(G)$ is then affine, and therefore the morphism $G \rightarrow G/\underline{\mathrm{Centr}}(G)$ is affine. Since the latter group is affine (Exposé XII, 6.1),¹¹ G is affine.

When G is reductive and T is maximal, the roots in the preceding sense coincide with the roots in the sense of *Bible*, §12.2, Definition 1. The latter are indeed roots in the sense of this number (*Bible*, §13.2, Theorem 1(c)), and their number is $\dim(G) - \dim(T)$ (*Bible*, §13.4, Corollary 2 to Theorem 3). Moreover, if α is a root, then so is $-\alpha$ (*Bible*, §12.2, Corollary to Proposition 1). As usual, we write the group law of $\mathrm{Hom}_{k\text{-gr}}(T, \mathbb{G}_{m,k})$ indifferently additively or multiplicatively. It follows that, for reductive G ,

$$(1.10.4) \quad \dim(G) - \mathrm{rgred}(G) = \mathrm{Card}(R)$$

is always even.

⁹Editor's note: the following sentence has been added.

¹⁰Editor's note: the following argument has been given in detail.

¹¹Editor's note: see also Addition VI_B, 6.2.6.

The semisimple rank of the reductive group G is the rank of the subset R of the $\mathbb{Q}$ -vector space

$$\mathrm{Hom}_{k\text{-gr}}(T, \mathbb{G}_{m,k}) \otimes_{\mathbb{Z}} \mathbb{Q}$$

(*Bible*, §13.3, Theorem 2).

Lemma 1.11.— *Let k be an algebraically closed field, G a smooth affine connected k -group, and T a torus of G . Put*

$$W(T) = \underline{\mathrm{Norm}}_G(T) / \underline{\mathrm{Centr}}_G(T),$$

the Weyl group of G relative to T . The following conditions are equivalent:

- (i) *G is reductive, T is maximal, and $\mathrm{rgss}(G) = 1$.*
- (ii) *G is reductive, T is maximal, and $G \neq T$; there exists a subtorus Q of T , of codimension 1 in T , that is central in G .*
- (iii) *G is not solvable and $\dim(G) - \dim(T) \leq 2$.*
- (iv) *$W(T) \neq \{e\}$ and $\dim(G) - \dim(T) \leq 2$.*
- (v) *$W(T) = (\mathbb{Z}/2\mathbb{Z})_k$ and $\dim(G) - \dim(T) = 2$.*

Moreover, under these conditions there are exactly two roots of G relative to T , and they are opposite. Under the conditions of (ii), $Q = \mathrm{rad}(G)$.

Proof. The implication (v) $\Rightarrow$ (iv) is clear, and (iv) $\Rightarrow$ (iii) follows from *Bible*, §6.1, Corollary 3 to Theorem 1, or [Ch05], §6.2, Corollary 3 to Theorem 2.

Let us prove (iii) $\Rightarrow$ (ii). Let U be the unipotent radical of G . The group G/U is reductive and is not a torus (otherwise G would be solvable); hence, by (1.10.4),

$$\dim(G/U) - \mathrm{rgred}(G/U) = \mathrm{Card}(R) \geq 2.$$

On the other hand,

$$\mathrm{rgred}(G) = \mathrm{rgred}(G/U) + \mathrm{rgred}(U) = \mathrm{rgred}(G/U),$$

and therefore

$$\begin{aligned} \dim(G) - \dim(U) - \mathrm{rgred}(G) &= \dim(G/U) - \mathrm{rgred}(G/U) \\ &\geq 2 \geq \dim(G) - \dim(T) \geq \dim(G) - \mathrm{rgred}(G). \end{aligned}$$

It follows that $\dim(U) = 0$, so G is reductive, and that $\mathrm{rgred}(G) = \dim(T)$; thus T is maximal and $\dim(G) - \dim(T) = 2$. By *Bible*, §10.4, Proposition 8, there exists a singular torus Q of codimension 1 in T . Then $\underline{\mathrm{Centr}}_G(Q)$ is reductive (1.6.2(i)) and not solvable (by the definition of a singular torus), and hence

$$\dim(\underline{\mathrm{Centr}}_G(Q)) - \mathrm{rgred}(G) \geq 2.$$

This proves that $G = \underline{\mathrm{Centr}}_G(Q)$, and that Q is central in G .

Let us prove (ii) $\Rightarrow$ (i). We know that G/Q is reductive by 1.7 and that $\mathrm{rgred}(G/Q) = 1$; hence $\mathrm{rgss}(G/Q)$ is either 0 or 1. The first case is impossible, since it would imply $\mathrm{rgss}(G) = 0$, and therefore $G = T$. Thus

$$\mathrm{rgred}(G/Q) = \mathrm{rgss}(G/Q) = 1,$$

which proves that G/Q is semisimple. Consequently Q is the radical of G and $\text{rgss}(G) = 1$. Let us finally prove (i) $\Rightarrow$ (v). If Q is the radical of G , then $\dim(T) - \dim(Q) = 1$, and Q is central in G . Hence $G = \text{Centr}_G(Q)$, which proves that Q is a singular torus. By *Bible*, §11.3, Theorem 2,

$$W_G(T) = (\mathbb{Z}/2\mathbb{Z})_k,$$

and, by *Bible*, §12.1, Lemma 1, $\dim(G) - \dim(T) = 2$. Thus G has at most two roots relative to T , while it has at least two, opposite to one another, by 1.10.

Proposition 1.12.— *Let k be an algebraically closed field, G a smooth connected k -group, T a torus of G , and R the set of roots of G relative to T . Write*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha, \quad \mathfrak{g}^\alpha \neq 0,$$

for the decomposition of $\mathfrak{g}$ under $\text{Ad}(T)$. For every $\alpha \in R$, put

$$T_\alpha = \text{Ker}(\alpha)^0, \quad Z_\alpha = \text{Centr}_G(T_\alpha).$$

¹² The following conditions are equivalent:

- (i) G is affine and reductive, and T is maximal.
- (ii) $\mathfrak{g}^0 = \mathfrak{t}$, and every Z_α , for $\alpha \in R$, is reductive.
- (iii) $\mathfrak{g}^0 = \mathfrak{t}$; every $\mathfrak{g}^\alpha$, for $\alpha \in R$, has dimension 1; if $\alpha, \beta \in R$ and $\beta = q\alpha$ for some $q \in \mathbb{Q}$, then $q = \pm 1$; and, for every $\alpha \in R$, there exists $w_\alpha \in G(k)$ which normalizes T , centralizes T_α , but does not centralize T .

Moreover, under these conditions every Z_α has semisimple rank 1, and

$$\text{Lie}(Z_\alpha) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}.$$

Proof. Suppose first that G is affine and reductive and that T is maximal. Every Z_α is then affine and reductive by 1.6.2(i), with maximal torus T ; furthermore, T is its own centralizer by 1.6.2(ii). Thus $\mathfrak{g}^0 = \mathfrak{t}$, proving (i) $\Rightarrow$ (ii).

Conversely, the equality $\mathfrak{g}^0 = \mathfrak{t}$ implies that T is maximal and that G is affine, by 1.10. Hence every Z_α is affine, smooth, and connected, by 1.3. In all cases, Exposé II, 5.2.2 gives

$$\text{Lie}(Z_\alpha) = \mathfrak{g}^{T_\alpha} = \mathfrak{g}^0 \oplus \bigoplus_{\beta \in R \cap \mathbb{Q}\alpha} \mathfrak{g}^\beta.$$

Consequently,

$$\text{Lie}(Z_\alpha) \supset \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

so $Z_\alpha \neq T$. Since T_α is a central subtorus of Z_α , of codimension 1 in T , Lemma 1.11 applied to Z_α proves the equivalence of (ii) and (iii), together with the additional assertions.

It remains to prove (ii) $\Rightarrow$ (i). We already know that (ii) implies that T is maximal and that G is affine. Let U be the unipotent radical of G . It is invariant in G , and its Lie algebra is invariant under $\text{Ad}(T)$; hence

$$\text{Lie}(U) = (\text{Lie}(U) \cap \mathfrak{g}^0) \oplus \bigoplus_{\alpha \in R} (\text{Lie}(U) \cap \mathfrak{g}^\alpha).$$

¹²Editor's note: thus T_α is the identity component of $\text{Ker}(\alpha)$.

By (1.3.1),

$$\begin{aligned} U \cap T &= U \cap \underline{\text{Centr}}_G(T) = \text{rad}^u(\underline{\text{Centr}}_G(T)) = \text{rad}^u(T) = \{e\}, \\ U \cap Z_\alpha &= U \cap \underline{\text{Centr}}_G(T_\alpha) = \text{rad}^u(\underline{\text{Centr}}_G(T_\alpha)) = \text{rad}^u(Z_\alpha) = \{e\}. \end{aligned}$$

It follows¹³ that

$$\begin{aligned} \text{Lie}(U) \cap \mathfrak{g}^0 &= \text{Lie}(U) \cap \mathfrak{t} = \text{Lie}(U \cap T) = 0, \\ \text{Lie}(U) \cap \mathfrak{g}^\alpha &\subset \text{Lie}(U) \cap \text{Lie}(Z_\alpha) = \text{Lie}(U \cap Z_\alpha) = 0. \end{aligned}$$

Thus $\text{Lie}(U) = 0$, and therefore $U = \{e\}$; in other words, G is reductive.

Corollary 1.13.— *Let k , G , T , R , and*

$$\mathfrak{g} = \mathfrak{g}^0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha$$

be as in Proposition 1.12. For every $\alpha \in R$, let T_α be the maximal torus of $\text{Ker}(\alpha)$ and put $Z_\alpha = \underline{\text{Centr}}_G(T_\alpha)$. The following conditions are equivalent:

- (i) *G is affine and semisimple, and T is maximal.*
- (ii) *$\mathfrak{g}^0 = \mathfrak{t}$, every Z_α is reductive, and $\bigcap_{\alpha \in R} \text{Ker}(\alpha)$ is finite.*

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2. Reductive group schemes: definitions and first properties

Scholium 2.1.— *For an S -group scheme G , the following properties are equivalent:*

- (i) *G is affine and smooth over S , with connected fibers.*
- (ii) *G is affine and flat over S , of finite presentation over S , with geometrically integral fibers.*

These properties are stable under base change and are local for the fpqc topology.

¹⁵ Indeed, suppose that (i) holds. Since G is affine and smooth over S , it is of finite presentation over S ; and since its fibers are smooth and connected, they are geometrically integral, by Exposé VI_A, 2.4.

Conversely, if (ii) holds, the fibers of G are connected and geometrically reduced, and hence smooth (Exposé VI_A, 1.3.1). It follows that G is smooth over S , by EGA IV₄, 17.5.2.

These properties are plainly stable under base change: see EGA II, 1.5.1 for “affine”, EGA IV₁, 1.6.2(iii) for “of finite presentation”, EGA IV₂, 2.1.4 for “flat”, and EGA IV₂, 4.6.5(i) for “with geometrically integral fibers”.

They are also clearly local for the Zariski topology. It is therefore enough to check that, if $S' \rightarrow S$ is faithfully flat and quasi-compact and $G_{S'} \rightarrow S'$ has the indicated properties, then so does $G \rightarrow S$. This follows from EGA IV₂, 2.5.1 for “flat”, 2.7.1(vi) and (xiii) for “of finite presentation” and “affine”, and 4.6.5(i) for “with geometrically integral fibers”; for “smooth”, see also EGA IV₄, 17.7.3(ii).

¹³Editor’s note: if U and L are two subgroup schemes of G , then $\text{Lie}(U) \cap \text{Lie}(L) = \text{Lie}(U \cap L)$.

¹⁴Editor’s note: this follows from the equalities $\text{rad}(G) = \underline{\text{Centr}}(G)^0$ and $\underline{\text{Centr}}(G) = \bigcap_{\alpha \in R} \text{Ker}(\alpha)$, established in 1.6.2(iv) and (1.10.3).

¹⁵Editor’s note: the following details have been added.

2.2. Let G be an S -group scheme satisfying the preceding conditions, and let Q be a torus of G in the sense of Exposé IX, Definition 1.3.¹⁶ By Exposé XI, 6.11(a) and 2.4, $\underline{\text{Centr}}_G(Q)$ is represented by a closed subgroup scheme of G ; it is therefore affine over S , and it is of finite presentation and smooth over S . Moreover, every geometric fiber $G_{\bar{s}}$ is a smooth affine connected $\kappa(\bar{s})$ -group. The first assertion of 1.3 therefore shows that the same is true of

$$\underline{\text{Centr}}_{G_{\bar{s}}}(Q_{\bar{s}}) = (\underline{\text{Centr}}_G(Q))_{\bar{s}}.$$

Lemma 2.3.— *Let S be a scheme, let G be an S -group scheme that is smooth and affine over S , with connected fibers, and let T be a torus of G . The set of points $s \in S$ for which $G_{\bar{s}}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}}$, is an open subset U of S .*

Proof. Since G and T are flat over S , the function

$$s \longmapsto \dim(G_{\bar{s}}/\bar{s}) - \dim(T_{\bar{s}}/\bar{s})$$

is locally constant. Let U_1 be the open subset of S on which this function has the value 2.

¹⁷ By 6.3 the Weyl group

$$W_G(T) = \underline{\text{Norm}}_G(T)/\underline{\text{Centr}}_G(T)$$

is represented by an S -group scheme that is étale and separated over S , and the function

$$s \longmapsto \text{the number of points of } W_G(T)_{\bar{s}}$$

is lower semicontinuous. Let U_2 be the locus on which this number is greater than 1; it is open. By Lemma 1.11, the locus on which $G_{\bar{s}}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}}$, is

$$U = U_1 \cap U_2.$$

Furthermore, $W_G(T)_{\bar{s}}$ has exactly two points for every $s \in U$. Consequently, by SGA 1, I, 10.9 and EGA IV₃, 15.5.1 and IV₄, 18.12.4, $W_G(T)_U$ is finite and étale over U .

Remark 2.4.— *The group $W_G(T)_U$ is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_U$.*

Indeed, the preceding argument shows that it is finite and étale over U . Since the automorphism functor of $(\mathbb{Z}/2\mathbb{Z})_U$ is the trivial group, it is enough to verify the assertion locally. It is immediate whenever the algebra $\mathcal{A}$ defining $W_G(T)_U$ is a free $\mathcal{O}_U$ -module.¹⁸

Notations and reminders 2.5.0.—¹⁹ Let S be a scheme, let G be an S -group scheme, and let $\epsilon: S \rightarrow G$ be the unit section of G . We saw in II, §4.11 that the functor $\underline{\text{Lie}}(G/S)$ is represented by the vector bundle $\mathbb{V}(\omega_{G/S}^1)$, where $\omega_{G/S}^1 = \epsilon^*(\Omega_{G/S}^1)$, and we write

$$\mathfrak{g} = \underline{\text{Lie}}(G/S) = (\omega_{G/S}^1)^\vee$$

¹⁶Editor's note: the following argument has been given in detail.

¹⁷Editor's note: the following details have been added. Note also that §6 is independent of the rest of this Exposé.

¹⁸Editor's note: the hypothesis implies that $\mathcal{A}$ is locally free of rank 2. Since the augmentation ideal $\mathcal{J}$ is a direct summand of $\mathcal{A}$, after replacing U by a sufficiently small affine open $\text{Spec}(R)$, one may suppose that $I = \Gamma(U, \mathcal{J})$ is a free R -module of rank 1. If x is a generator of I , then $x^2 = ax$ for some $a \in R$; the assumption that $A = \Gamma(U, \mathcal{A})$ is étale implies that a is invertible, which gives the asserted local identification.

¹⁹Editor's note: this paragraph of notations and reminders has been added.

for the sheaf of sections of this vector bundle. Suppose in addition that G is smooth over S . Then $\omega_{G/S}^1$, and hence $\mathrm{Lie}(G/S)$, are finite locally free $\mathcal{O}_S$ -modules, and (cf. I, 4.6.5.1)

$$\underline{\mathrm{Lie}}(G/S) = \mathbb{W}(\mathrm{Lie}(G/S));$$

that is, for every S -scheme S' ,

$$\underline{\mathrm{Lie}}(G/S)(S') = \Gamma(S', \mathrm{Lie}(G/S) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}).$$

By II, 4.1.1.1, the adjoint action of G gives $\underline{\mathrm{Lie}}(G/S) = \mathbb{W}(\mathrm{Lie}(G/S))$ the structure of a G - $\mathcal{O}_S$ -module; thus $\mathrm{Lie}(G/S)$ is a G - $\mathcal{O}_S$ -module (cf. I, 4.7.1). If G is moreover affine over S , this says, by I, 4.7.2, that $\mathrm{Lie}(G/S)$ is an $A(G)$ -comodule.

If T is a torus over S (IX, Definition 1.3), we say that T is *split* (“trivial” in the original) if it is isomorphic to $\mathbb{G}_{m,S}^r$ for some integer $r \geq 0$, and that T is *trivialized* if such an isomorphism has been fixed, or more generally if an isomorphism

$$T \simeq D_S(M)$$

has been fixed, where M is a free abelian group of rank r .

Finally, recall (XII, Definition 1.3) that a torus T of G is called *maximal* if, for every $s \in S$, $T_{\bar{s}}$ is a maximal torus of $G_{\bar{s}}$.

Theorem 2.5.— *Let S be a scheme, let G be an S -group scheme that is affine and of finite presentation over S , with connected fibers, and let s_0 be a point of S . Suppose that G is flat over S at $\epsilon(s_0)$, and that the geometric fiber $G_{\bar{s}_0}$ is (reduced and) reductive (resp. semisimple). Then there exist an open subset $U \subset S$, containing s_0 , and an étale surjective morphism $S' \rightarrow U$ such that:*

- (i) G_U is smooth, with reductive (resp. semisimple) geometric fibers, and its reductive rank and semisimple rank are constant.
- (ii) $G_{S'}$ has a split²⁰ maximal torus T , and the Weyl group (cf. 6.3)

$$W_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T) / \underline{\mathrm{Centr}}_{G_{S'}}(T) = \underline{\mathrm{Norm}}_{G_{S'}}(T) / T$$

is finite over S' .

Proof. ²¹ Write $\pi: G \rightarrow S$ for the structure morphism and $\epsilon: S \rightarrow G$ for the unit section. Since G is flat over S at $\epsilon(s_0)$, and since $G_{\bar{s}_0}$ is reduced (hence smooth over $\bar{s}_0$, by VI_A, 1.3.1), G is smooth over S at $\epsilon(s_0)$ (EGA IV₄, 17.5.1). Thus there is an open neighborhood V of $\epsilon(s_0)$ on which π is smooth. Then $S' = \epsilon^{-1}(V)$ is open in S , and $G_{S'}$ is smooth over S' at the points of $\epsilon(S')$. Since G has connected fibers, VI_B, 3.10 shows that $G_{S'}$ is smooth over S' . Replacing S by S' , we may therefore suppose that G is smooth over S .

By Exposé XI, Theorem 4.1, the functor of subgroups of multiplicative type of G is represented by an S -scheme $\mathcal{M}$, smooth and separated over S . Let r_0 be the rank of $G_{\bar{s}_0}$, and consider the open subscheme $\mathcal{M}_{r_0} \subset \mathcal{M}$ representing the functor of subtori of rank r_0 of G (IX, 1.4). Smoothness implies that $\mathcal{M}_{r_0}$ has a rational point over a finite separable extension of $\kappa(s_0)$ (EGA IV₄, 17.15.10(iii)). Consequently there is an étale morphism $S' \rightarrow S$, together with a point s'_0 above s_0 , such that $G_{s'_0}$ has a torus of rank r_0 . Replacing S by S' , we may therefore suppose that G_{s_0} has a torus of rank r_0 .

²⁰Editor’s note: the word “split” has been added.

²¹Editor’s note: the proof has been expanded.

By “Hensel’s lemma” (XI, 1.10), the smoothness of $\mathcal{M}_{r_0}$ allows this torus to be lifted to an S' -torus T of G , where $S' \rightarrow S$ is étale and is equipped with a point s'_0 above s_0 . By Exposé X, 4.5 (see also 6.1²²), there is an étale morphism $f: S'' \rightarrow S'$ that splits T and for which $f^{-1}(s'_0) \neq \emptyset$. Because an étale morphism is open, and the assertions in (i) are local for the étale topology, we may thus suppose that G has a split torus T ,²³ maximal at s_0 . Write $T = D_S(M)$, and let

$$\mathfrak{g} = \prod_{m \in M} \mathfrak{g}^m$$

be the decomposition of $\mathfrak{g} = \text{Lie}(G/S)$ under $\text{Ad}(T)$ (Exposé I, 4.7.3).

Put $\mathfrak{t} = \text{Lie}(T)$, and, for every $m \in M$, write

$$\mathfrak{g}^m(s_0) = \mathfrak{g}^m \otimes_{\mathcal{O}_S} \kappa(s_0).$$

²⁴ Let R be the set of $m \in M$, $m \neq 0$, such that $\mathfrak{g}^m(s_0) \neq 0$.²⁵ Since $G_{\bar{s}_0}$ is reductive, $\mathfrak{g}^0(s_0) = \mathfrak{t}(s_0)$, and hence

$$\mathfrak{g}(s_0) = \mathfrak{t}(s_0) \oplus \prod_{\alpha \in R} \mathfrak{g}^\alpha(s_0).$$

Since the modules in question are locally free, after shrinking S we may suppose that the $\mathfrak{g}^\alpha$ are free and that

$$\mathfrak{g} = \mathfrak{t} \oplus \prod_{\alpha \in R} \mathfrak{g}^\alpha.$$

Recall from Exposé XII, 1.12 that a group of multiplicative type has a unique maximal torus (this also follows almost immediately by descent, the diagonalizable case being evident). Let T_α be the maximal torus of $\ker(\alpha)$, and put

$$Z_\alpha = \underline{\text{Centr}}_G(T_\alpha).$$

²⁶ By 2.2, Z_α is affine and smooth over S , with connected fibers; and by 1.12, its fiber $(Z_\alpha)_{\bar{s}_0}$ is reductive, of semisimple rank 1, with maximal torus $T_{\bar{s}_0}$. By 2.3 there is therefore an open subset $U_\alpha \subset S$, containing s_0 , on which Z_α has reductive fibers.

Set

$$U = \bigcap_{\alpha \in R} U_\alpha.$$

By 1.12, for every $s \in U$, $G_{\bar{s}}$ is reductive, with maximal torus $T_{\bar{s}}$, and the set of roots of $G_{\bar{s}}$ relative to $T_{\bar{s}}$ is identified with R . Consequently

$$\text{rgred}(G_{\bar{s}}) = \dim(T) = \text{rg}(M), \quad \text{rgss}(G_{\bar{s}}) = \text{rg}(R) \quad (\text{cf. 1.10}).$$

This proves (i) and the first assertion of (ii). It remains to prove that $W_{G_U}(T_U)$ is finite over U , that is, that it has the same number of points in every geometric fiber (SGA 1, I, 10.9; EGA IV₃, 15.5.1; and EGA IV₄, 8.12.4).

For this it is enough to observe that, for $s \in U$, the geometric fiber of this group is determined by the configuration $R \subset M$: it is the constant group associated with the abstract Weyl group of this root system. In particular it is independent of s ; see the *Bible*, §11.3, Theorem 2 (and also Exposé XXII, no. 3). $\square$

²²Editor’s note: §6.1 is independent of the rest of this Exposé.

²³Editor’s note: throughout this volume, “trivial torus” has been replaced by “split torus”.

²⁴Editor’s note: the preceding sentence has been added.

²⁵Editor’s note: since $\mathfrak{g}$ is a finite locally free $\mathcal{O}_S$ -module, R is finite.

²⁶Editor’s note: the following argument has been expanded.

Corollary 2.6.— *Let G be an S -group scheme that is affine and smooth over S , with connected fibers. The set of points $s \in S$ for which $G_{\bar{s}}$ is reductive (resp. semisimple) is an open subset U of S , and the functions*

$$s \mapsto \operatorname{rgred}(G_{\bar{s}}/\bar{s}), \quad s \mapsto \operatorname{rgss}(G_{\bar{s}}/\bar{s})$$

are locally constant on U .

Definition 2.7.— *An S -group (that is, an S -group scheme) G is called reductive (resp. semisimple) if it is affine and smooth over S , with connected reductive (resp. semisimple) geometric fibers.*

For an S -group scheme G , the property of being reductive (resp. semisimple) is stable under base change and local for the fpqc topology.

2.8.— Let G be a reductive S -group. For every torus (resp. maximal torus) Q of G ,

$$Z(Q) = \operatorname{Centr}_G(Q)$$

is reductive (resp. is equal to Q). This follows at once from 2.2 and 1.6. Applying 2.5 to $\operatorname{Centr}_G(Q)$, we deduce that Q is contained, locally for the étale topology, in a maximal torus.

Remark 2.9.— *Using, as usual, the technique of EGA IV₃, §8, the finite-presentation hypotheses, and Theorem 2.5, one sees that if G is a reductive group over S , there is an open cover $\{U_i\}$ of S such that each G_{U_i} is obtained by base change from a reductive group over a noetherian ring (in fact, over a finitely generated $\mathbb{Z}$ -algebra). Likewise, if G has a split maximal torus T , one may further suppose that T_{U_i} comes from a split maximal torus of the preceding group, ...*

3. Roots and root systems of reductive group schemes

3.1.— Let S be a scheme, and let T be an S -torus acting linearly on a finite locally free $\mathcal{O}_S$ -module F (cf. I, §4.7). For every character α of T , that is,

$$\alpha \in \operatorname{Hom}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}),$$

define a subfunctor of $\mathbb{W}(F)$ by

$$\mathbb{W}(F)^\alpha(S') = \{x \in \mathbb{W}(F)(S') \mid t \cdot x = \alpha(t)x \text{ for every } t \in T(S''), S'' \rightarrow S'\}.$$

Lemma.—²⁷ *Then $\mathbb{W}(F)^\alpha = \mathbb{W}(F^\alpha)$, where F^α is a submodule of F , locally a direct summand of F , and hence also locally free.*

Indeed, the assertion is local for the fpqc topology, and we may suppose that $T = D_S(M)$, where M is a finite-type (free) abelian group. Then α is identified with a locally constant function from S to M (Exposé VIII, 1.3), and, after restricting S if necessary, we may suppose that this function is constant. We are then reduced to Exposé I, 4.7.3.

Definition 3.2.— *Let S be a scheme, let G be an S -group scheme smooth over S and with connected fibers,²⁸ and let T be a torus of G . Put $\mathfrak{g} = \operatorname{Lie}(G/S)$, and let T act on $\mathfrak{g}$ through the adjoint representation of G .*

²⁷Editor's note: this statement has been made into a lemma in order to bring it out.

²⁸Editor's note: the original also assumed that G was of finite presentation over S , an assumption that does not appear to be used below. In any case, since G is smooth over S and has connected fibers, it is quasi-compact and separated over S (VI_B, 5.5), and hence of finite presentation over S .

A character α of T is called a root of G with respect to T if the following equivalent conditions are satisfied:

- (i) For every $s \in S$, α_s is a root of G_s with respect to T_s (1.10).
- (ii) α is nontrivial on every fiber, and $\mathfrak{g}^\alpha(s) \neq 0$ for every $s \in S$.

Lemma 3.3.— Let S be a scheme, let T be an S -torus, and let α be a character of T . The following conditions are equivalent:

- (i) α is nontrivial on every fiber; that is, for every $s \in S$, α_s is distinct from the unit character of T_s .
- (ii) For every $S' \rightarrow S$ with $S' \neq \emptyset$, $\alpha_{S'}$ is distinct from the unit character of $T_{S'}$.
- (iii) The morphism α is faithfully flat.

²⁹ It is clear that (ii) implies (i), and it is easy to see that (iii) implies (i). Condition (i) implies (ii): if $s' \in S'$ lies over s , and $\alpha_{s'}$ is the unit character, then so is α_s . Finally, to prove that (i) implies (iii), we reduce to the case in which T is diagonalizable and conclude by Exposé VIII, 3.2(a).

3.4.—³⁰ Let G be a reductive S -group scheme, let T be a maximal torus of G , and let α be a root of G with respect to T . Then, by 2.5.0 and 1.12, $\mathfrak{g}^\alpha$ is a locally free $\mathcal{O}_S$ -module of rank one. Moreover, by 1.10, $-\alpha$ is also a root of G with respect to T . In particular, if G has semisimple rank one, 1.11 and 1.12 give the following statement.

Lemma 3.5.— Let S be a scheme, let G be a reductive S -group of semisimple rank one, and let T be a maximal torus of G . If α is a root of G with respect to T , then so is $-\alpha$, and

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are locally free of rank one.

Definition 3.6.— Let S be a scheme, let G be a reductive S -group, let T be a maximal torus of G , and let R be a set of roots of G with respect to T . We say that R is a root system of G with respect to T if the following equivalent conditions are satisfied:

- (i) For every $s \in S$, the map $\alpha \mapsto \alpha_s$ is a bijection from R onto the set of roots of G_s with respect to T_s .
- (ii) The elements of R are distinct on every fiber (that is, if $\alpha, \alpha' \in R$ and $\alpha \neq \alpha'$, then $\alpha - \alpha' = \alpha\alpha'^{-1}$ is nontrivial on every fiber), and, for every $s \in S$,

$$\dim(G_s/\kappa(s)) - \dim(T_s/\kappa(s)) = \text{Card}(R).$$

- (iii) One has

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}^\alpha.$$

²⁹Editor's note: the original has been simplified; it invoked Exposé IX, 5.2. Note, however, that loc. cit., 5.3 shows that, if S is connected, the conditions of the lemma hold as soon as (i) holds at one point s of S .

³⁰Editor's note: what follows has been modified so as to recall the hypotheses of 3.2 and to add that T is a maximal torus.

The equivalence of these conditions is immediate.

Lemma 3.7.— *Let S be a scheme, let G be a reductive S -group, let T be a maximal torus of G , and let R be a root system of G with respect to T . Every root of G with respect to T is locally on S equal to an element of R .*

Proof. This follows from condition (iii).

Put

$$\mathcal{M} = \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S});$$

this is a twisted constant S -group scheme (Exposé X, 5.6). If G has a root system R with respect to T , then the inclusion $R \hookrightarrow \mathcal{M}(S)$ defines a morphism

$$R_S \longrightarrow \mathcal{M},$$

where R_S is the constant S -scheme defined by R . By condition (ii), this morphism is an open and closed immersion whose image is precisely

$$\bigcup_{\alpha \in R} \alpha(S),$$

where each $\alpha \in R$ is regarded as a section $S \rightarrow \mathcal{M}$.

Let $\mathcal{R}$ be the functor of roots of G with respect to T . By definition, $\mathcal{R}(S')$ is the set of roots of $G_{S'}$ with respect to $T_{S'}$, for every $S' \rightarrow S$. If $S' = \emptyset$, set $\mathcal{R}(\emptyset) = \{e\}$; if $S' \neq \emptyset$, the inclusion $R \hookrightarrow \mathcal{M}(S')$ identifies R with a root system of $G_{S'}$ with respect to $T_{S'}$. Thus, by 3.7,

$$\mathcal{R}(S') = \mathrm{Hom}_{\mathrm{loc. const.}}(S', R),$$

which shows that $\mathcal{R}$ is represented by R_S .

Now drop the assumption that G necessarily has a root system with respect to T . The functor $\mathcal{R}$ is in any case an fpqc subsheaf of $\mathcal{M}$. Locally for this topology, G does have a root system with respect to T (for example, make T split and use the proof of Theorem 2.5). By Exposé IV, 4.6.8 and the descent theory of open (respectively closed) subschemes,³¹ we obtain the following proposition.

Proposition 3.8.— *Let S be a scheme, let G be an S -group, and let T be a maximal torus of G . The functor $\mathcal{R}$ of roots of G with respect to T is represented by a finite twisted constant S -scheme (Exposé X, 5.1) which is an open and closed subscheme of*

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}).$$

A subset $R \subset \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S})$ is a root system of G with respect to T if and only if the morphism

$$R_S \longrightarrow \underline{\mathrm{Hom}}_{S\text{-gr.}}(T, \mathbb{G}_{m,S}),$$

defined by the preceding inclusion, induces an isomorphism

$$R_S \xrightarrow{\sim} \mathcal{R}.$$

3.9.— Let S be a scheme, let G be a reductive S -group, let T be a maximal torus of G , and let α be a root of G with respect to T , that is, a section of $\mathcal{R}$. Consider the kernel $\mathrm{Ker}(\alpha)$ of α , its unique maximal torus T_α , and the centralizer of the latter,

$$Z_\alpha = \underline{\mathrm{Centr}}_G(T_\alpha).$$

³¹Editor's note: see SGA 1, VIII, 4.4 or EGA IV₂, 2.7.1.

This is a closed S -subgroup of G , reductive (2.8), and of semisimple rank one (1.12). Moreover,

$$\mathrm{Lie}(Z_\alpha/S) = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha};$$

hence $\{\alpha, -\alpha\}$ is a root system of Z_α with respect to T .

3.10.— Let S be a scheme, let G be a reductive S -group, let T be a maximal torus of G , and let α be a root of G with respect to T . If $q \in \mathbb{Q}$ and $q\alpha$ is a root of G with respect to T , then $q = 1$ or $q = -1$. This follows at once from 1.12.

4. Roots and vector group schemes

4.1.— Let S be a scheme and let F be a finite locally free $\mathcal{O}_S$ -module. The S -scheme $\mathbb{W}(F)$ is smooth over S . Its Lie algebra is canonically isomorphic to F . Indeed, there is a canonical isomorphism

$$\mathbb{W}(F) \xrightarrow{\sim} \mathrm{Lie}(\mathbb{W}(F)/S) = \mathbb{W}(\mathrm{Lie}(\mathbb{W}(F)/S))$$

(Exposé II, 4.4.1 and 4.4.2). We shall always identify F with $\mathrm{Lie}(\mathbb{W}(F)/S)$.

Lemma 4.2.— *Let S be a scheme and let V be a vector bundle over S , smooth over S . There is then a unique isomorphism of $\mathcal{O}_S$ -modules*

$$\exp: \mathbb{W}(\mathrm{Lie}(V/S)) \xrightarrow{\sim} V$$

which induces the identity on Lie algebras.³²

Proof. We have $V = \mathbb{V}(\mathcal{F})$ for some quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. Since V is smooth over S , the module $\mathcal{F} \simeq \omega_{V/S}^1$ is finite locally free, and hence

$$\mathrm{Lie}(V/S) = \mathbb{V}(\omega_{V/S}^1) \simeq \mathbb{W}(\mathrm{Lie}(V/S)).$$

³²Editor's note: the proof in the original edition has been retained; it may also be expanded as follows.

Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$ -module and put $V = \mathbb{V}(\mathcal{F})$. Write $\pi: V \rightarrow S$ for the projection and $\epsilon: S \rightarrow V$ for the zero section. Then $\Omega_{V/S}^1 = \pi^* \mathcal{F}$, whence

$$(1) \quad \omega_{V/S}^1 = \epsilon^* \Omega_{V/S}^1 \simeq \epsilon^* \pi^* \mathcal{F} \simeq \mathcal{F},$$

and therefore $\mathrm{Lie}(V/S) = (\omega_{V/S}^1)^\vee \simeq \mathcal{F}^\vee$. If V is smooth over S , then by (1) $\mathcal{F}$ is finite locally free, and hence

$$(2) \quad V = \mathbb{V}(\mathcal{F}) \simeq \mathbb{W}(\mathcal{F}^\vee) \simeq \mathbb{W}(\mathrm{Lie}(V/S)).$$

Now let U be an S -scheme equipped with a section $\tau: S \rightarrow U$, and suppose that an isomorphism $\phi: V \xrightarrow{\sim} U$ of pointed S -schemes has been given, that is, $\phi \circ \epsilon = \tau$ (for example, when U is an S -group). Then ϕ induces an isomorphism

$$d\phi: \mathcal{F}^\vee \xrightarrow{\sim} \mathrm{Lie}(U/S),$$

and therefore an isomorphism ψ displayed by

$$\begin{array}{ccccc} \mathbb{W}(\mathcal{F}^\vee) & \xlongequal{\quad} & \mathbb{V}(\mathcal{F}) & \xrightarrow[\sim]{\phi} & U \\ & \nwarrow \scriptstyle \mathbb{W}(d\phi^{-1}) & & \nearrow \scriptstyle \sim & \\ & & \mathbb{W}(\mathrm{Lie}(U/S)) & & \end{array}$$

which identifies U with $\mathbb{W}(\mathrm{Lie}(U/S))$. Since the functor $\mathbb{W}$ is fully faithful, there is a unique isomorphism

$$\exp: \mathbb{W}(\mathrm{Lie}(U/S)) \xrightarrow{\sim} U$$

such that $\mathrm{Lie}(\psi^{-1} \circ \exp) = \mathrm{id}$. In fact, one sees directly that $\mathrm{Lie}(\psi) = \mathrm{id}$, so that $\exp = \psi$.

Moreover, by Exposé II, loc. cit., there is a canonical isomorphism

$$V \xrightarrow{\sim} \mathrm{Lie}(V/S) \simeq \mathbb{W}(\mathrm{Lie}(V/S)),$$

and the uniqueness of $\exp$ follows at once because $\mathbb{W}$ is fully faithful. $\square$

4.3. Notation.— If V is a vector bundle over S , let $V^\times$ denote the open subset obtained from V by removing the zero section. Write the group law of V multiplicatively. The action of $\mathcal{O}_S$ on V which defines its module structure will then be written exponentially:

$$\mathcal{O}_S \times_S V \longrightarrow V, \quad (x, v) \longmapsto v^x.$$

Thus

$$(vv')^x = v^x(v')^x, \quad v^{x+x'} = v^x v^{x'}, \quad v^{xx'} = (v^x)^{x'}.$$

In particular, upon restricting the operator action to $\mathbb{G}_{m,S}$, the open subset $V^\times$ is stable and is therefore an object with operator group $\mathbb{G}_{m,S}$. We shall continue to write this action as $(z, v) \longmapsto v^z$.

Definition 4.4.1.³³— Let L be an invertible module on S , and let $\mathbb{W}(L)$ be the associated vector bundle. Then $\mathbb{W}(L)^\times$ is a (locally trivial) principal homogeneous bundle under $\mathbb{G}_{m,S}$. We write

$$\Gamma(S, L)^\times = \mathbb{W}(L)^\times(S).$$

Scholium 4.4.2.— There are natural bijections among the isomorphisms of $\mathcal{O}_S$ -modules $\mathcal{O}_S \simeq L$, the isomorphisms of $\mathcal{O}_S$ -modules $\mathcal{O}_S \simeq \mathbb{W}(L)$,³⁴ and the sections $S \rightarrow \mathbb{W}(L)^\times$.

The correspondence is

$$f \longmapsto \mathbb{W}(f) \longmapsto \mathbb{W}(f)(1).$$

It is compatible with base change. Thus $\mathbb{W}(L)^\times$ may be regarded as the “scheme of trivializations of $\mathbb{W}(L)$ ”.

4.5.— Let S be a scheme, let T be an S -torus, let P be an S -group with operator group $\mathbb{G}_{m,S}$ (for example, a vector bundle), and let α be a character of T . We write $T \cdot_\alpha P$ for the semidirect product of P by T , where T acts on P through the composite morphism

$$T \xrightarrow{\alpha} \mathbb{G}_{m,S} \longrightarrow \underline{\mathrm{Aut}}_{S\text{-gr.}}(P).$$

Definition 4.6.— Let S be a scheme, let G be an S -group scheme, let T be a subgroup of G , let α be a character of T , and let L be an $\mathcal{O}_S$ -module. Let

$$p: \mathbb{W}(L) \longrightarrow G$$

be a morphism of S -group functors.³⁵ We say that p is normalized by T with multiplier α if it satisfies the following equivalent conditions:

³³Editor’s note: for purposes of later references, paragraph 4.4 has been divided into 4.4.1 and 4.4.2.

³⁴Editor’s note: here the vector bundle $\mathbb{W}(L)$ is identified with the $\mathcal{O}_S$ -module functor which it represents.

³⁵Editor’s note: the edition uses a bold $\mathbf{W}(L)$ here because, for an arbitrary $\mathcal{O}_S$ -module L , the functor need not be representable. Below, however, L is assumed finite locally free (indeed, invertible), in which case the functor is represented by the vector bundle $\mathbb{W}(L^\vee)$, and the notation $\mathbb{W}(L)$ is used without this distinction.

- (i) p is a morphism of objects with operator group T , where T acts on $\mathbb{W}(L)$ through α , and on G by inner automorphisms. In other words, for every $S' \rightarrow S$, every $t \in T(S')$, and every

$$x \in \mathbb{W}(L)(S') = \Gamma(S', L \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}),$$

one has

$$\text{int}(t)p(x) = p(\alpha(t)x).$$

- (ii) The morphism

$$T \cdot_{\alpha} \mathbb{W}(L) \longrightarrow G, \quad (t, x) \longmapsto tp(x),$$

defined by multiplication in G , is a group morphism.

Lemma 4.7.— *Under the hypotheses of 4.6, suppose in addition that G is smooth with connected fibers,³⁶ that T is a maximal torus of G , and that L is invertible. If p is a monomorphism and α is nonzero on every fiber, then α is a root of G with respect to T .*

Indeed,

$$\text{Lie}(p): L \longrightarrow \mathfrak{g}$$

is a monomorphism which factors through $\mathfrak{g}^{\alpha}$.

Proposition 4.8.— *Under the hypotheses of 4.7, suppose that G is reductive and that p is a monomorphism. Then α is a root of G with respect to T , and $\text{Lie}(p)$ induces an isomorphism*

$$\text{Lie}(p): L \xrightarrow{\sim} \mathfrak{g}^{\alpha}.$$

Indeed, by 4.7 and the fact that $\mathfrak{g}^{\alpha}$ is invertible, it is enough to prove that α is nonzero on every fiber. Let $s \in S$ be such that $\alpha_{\bar{s}} = 0$ ($= 1$ in multiplicative notation). If X is a nonzero section of $L \otimes_S \kappa(\bar{s})$, then $p(X)$ is a unipotent element different from the identity in $G(\bar{s})$ which centralizes $T_{\bar{s}}$. This is impossible, since $T_{\bar{s}}$ is its own centralizer.

Corollary 4.9.— *Under the hypotheses of 4.8, there exists a monomorphism of groups with operator group T (that is, normalized by T with multiplier α)*

$$\mathbb{W}(\mathfrak{g}^{\alpha}) \longrightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^{\alpha} \rightarrow \mathfrak{g}$.

We shall soon see that 4.9 in fact holds whenever G is a reductive group and α is a root of G with respect to T , and that such a morphism is unique.

Reminder 4.10.— *Let k be an algebraically closed field, let G be a reductive k -group, let T be a maximal torus of G , and let α be a root of G with respect to T . There exists a monomorphism*

$$p: \mathbb{G}_{a,k} \longrightarrow G$$

normalized by T with multiplier α .

See the *Bible*, §13.1, Theorem 1.

4.11.— We finish this section with a technical result which will be useful to us. Let S be a scheme, and let L be an invertible $\mathcal{O}_S$ -module. Let $q > 0$ be an integer such that $x \longmapsto x^q$ is an endomorphism of the S -group $\mathbb{G}_{a,S}$. If $S \neq \emptyset$, then $q = 1$, or $q = p^n$, where

³⁶Editor's note: compare editor's note (28).

p is a prime number which vanishes on S . This follows at once from the elementary fact that the greatest common divisor of the binomial coefficients $\binom{q}{i}$, for $i \neq 0, q$, is p if $q = p^n$ with p prime, and is 1 otherwise.

The morphism defined by taking the q -th power,

$$L \longrightarrow L^{\otimes q},$$

is a morphism of sheaves of abelian groups. By base change it defines a morphism of S -group schemes

$$\mathbb{W}(L) \longrightarrow \mathbb{W}(L^{\otimes q}).$$

In particular, if L' is another invertible module and an $\mathcal{O}_S$ -linear morphism

$$h: L^{\otimes q} \longrightarrow L'$$

is given, one obtains a morphism of S -group schemes

$$\mathbb{W}(L) \longrightarrow \mathbb{W}(L'), \quad x \longmapsto h(x^q).$$

With this notation, we have the following result.

Proposition 4.12.— *Let S be a scheme, let T (resp. T') be an S -torus, let L (resp. L') be an invertible $\mathcal{O}_S$ -module, and let α (resp. α') be a character of T (resp. T'). Let $f: T \rightarrow T'$ be a group morphism and let*

$$g: \mathbb{W}(L) \longrightarrow \mathbb{W}(L')$$

be a morphism of S -schemes (not necessarily a morphism of S -groups) satisfying

$$(\star) \quad g(\alpha(t)x) = \alpha'(f(t))g(x)$$

for every $S' \rightarrow S$, every $x \in \mathbb{W}(L)(S')$, and every $t \in T(S')$. Let $s_0 \in S$ be such that $\alpha_{\bar{s}_0} \neq 0$.³⁷

- (a) *Suppose that g sends the zero section to the zero section and that, for every integer $n > 0$,*

$$(\alpha' \circ f)_{\bar{s}_0} \neq n\alpha_{\bar{s}_0}.$$

Then $g = 0$ in a neighborhood of s_0 .

- (b) *Suppose that g is a group morphism and that $g_{\bar{s}_0} \neq 0$. There then exist an open subset $U \subset S$ containing s_0 and an integer $q > 0$ such that $x \longmapsto x^q$ is an endomorphism of $\mathbb{G}_{a,U}$ and*

$$(\alpha' \circ f)_U = q\alpha_U.$$

³⁷Editor's note: the phrases “consider the semidirect product $T \cdot_{\alpha} \mathbb{W}(L)$, respectively $T' \cdot_{\alpha'} \mathbb{W}(L')$ ” have been deleted. On the other hand, in part (b) the hypothesis that g be a group morphism, together with $(\star)$, is equivalent to requiring

$$(t, x) \longmapsto (f(t), g(x))$$

to be a group morphism from $T \cdot_{\alpha} \mathbb{W}(L)$ to $T' \cdot_{\alpha'} \mathbb{W}(L')$.

(c) Suppose that

$$(\alpha' \circ f)_{\bar{s}_0} = q\alpha_{\bar{s}_0},$$

where $q > 0$ is an integer such that $x \mapsto x^q$ is an endomorphism of $\mathbb{G}_{a,S}$. There then exist an open subset $U \subset S$ containing s_0 and a unique morphism of $\mathcal{O}_S$ -modules

$$h: L^{\otimes q}|_U \longrightarrow L'|_U$$

such that g_U is the composite

$$\mathbb{W}(L)_U \xrightarrow{x \mapsto x^q} \mathbb{W}(L^{\otimes q})_U \xrightarrow{\mathbb{W}(h)} \mathbb{W}(L')_U.$$

Proof of Proposition 4.12. Let us prove (a). Since the conclusion is local on S , we may suppose that $\mathbb{W}(L) = \mathbb{W}(L') = \mathbb{G}_{a,S}$, and hence that g is given by a polynomial

$$g(X) = \sum_{n \geq 0} a_n X^n, \quad a_n \in \Gamma(S, \mathcal{O}_S).$$

Condition $(\star)$, which relates f and g , is the identity in $\Gamma(S', \mathcal{O}_{S'})[X]$

$$\sum_{n \geq 0} a_n \alpha'(f(t)) X^n = \sum_{n \geq 0} a_n \alpha(t)^n X^n.$$

Thus, for every $n \geq 0$, every $S' \rightarrow S$, and every $t \in T(S')$,

$$a_n(\alpha'(f(t)) - \alpha(t)^n) = 0.$$

For every $n \geq 0$, let S_n be the set of points $s \in S$ such that

$$(\alpha' \circ f)_{\bar{s}} = n\alpha_{\bar{s}}.$$

It is known (Exposé IX, 5.3) that the S_n are both open and closed and that

$$(\alpha' \circ f)_{S_n} = n\alpha_{S_n}.$$

Moreover, since $\alpha_{\bar{s}_0} \neq 0$, after shrinking S we may suppose that α is nonzero on every fiber (same reference); it follows that the S_n are pairwise disjoint. After shrinking S once more, we may therefore suppose that one of the following two cases holds: $S = S_n$ for some n , or every S_n is empty.

Let $m \geq 0$ be such that $S_m = \emptyset$. We claim that $a_m = 0$. Indeed, $\alpha' \circ f$ and $m\alpha$ are distinct on every fiber of S , and we have:

Lemma 4.13.— *Let S be a scheme, let T be an S -torus, and let α, α' be two characters of T which are distinct on every fiber. There exist an fpqc covering family $\{S_i \rightarrow S\}$ and, for every i , an element $t_i \in T(S_i)$ such that*

$$\alpha(t_i) - \alpha'(t_i) = 1.$$

The lemma is immediate by reduction first to the diagonalizable case, and then to $T = \mathbb{G}_{m,S}$.

We return to the proof of Proposition 4.12. We have just proved (a). In cases (b) and (c), there is an integer n such that $S = S_n$ ($n = q$ in (c)). The preceding argument gives $a_m = 0$ for $m \neq n$, and therefore

$$g(X) = a_n X^n, \quad a_n \in \Gamma(S, \mathcal{O}_S).$$

This proves (c) at once. In case (b), we know that $a_n(s_0) \neq 0$; we may therefore suppose that a_n is invertible on S . It follows that $x \mapsto x^n$ is an endomorphism of $\mathbb{G}_{a,S}$, by the hypothesis in (b), and this completes the proof.

5. An instructive example

5.1.— Let k be an algebraically closed field of characteristic 0. Put $A = k[t]$, the polynomial ring in one variable over k , and let $S = \operatorname{Spec}(A)$. Consider the following Lie algebra $\mathfrak{g}$ over $\mathcal{O}_S$. As a module it is free of rank 3, with basis $\{X, Y, H\}$, and its multiplication table is

$$[X, Y] = 2tH, \quad [H, X] = X, \quad [H, Y] = -Y.$$

For $s \in S$, $s \neq s_0$, where s_0 is the point defined by $t = 0$, the fiber

$$\mathfrak{g}(s) = \mathfrak{g} \otimes_A \kappa(s)$$

is isomorphic to the Lie algebra of $\operatorname{PGL}_{2, \kappa(s)}$. For $s = s_0$, it is a solvable Lie algebra.

5.2.— Let G_1 be the group scheme of automorphisms of $\mathfrak{g}$: for every $S' \rightarrow S$, $G_1(S')$ is the group of automorphisms of the $\mathcal{O}_{S'}$ -Lie algebra $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$. It is a closed subscheme of the group $\operatorname{GL}(\mathfrak{g})$ of automorphisms of the $\mathcal{O}_S$ -module $\mathfrak{g}$.

Let $S' \rightarrow S$, and let

$$u \in M_3(\Gamma(S', \mathcal{O}_{S'}))$$

be viewed as an endomorphism of the $\mathcal{O}_{S'}$ -module $\mathfrak{g} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$:

$$\begin{aligned} u(X) &= aX + bY + eH, \\ u(Y) &= b'X + a'Y + e'H, \\ u(H) &= cX + c'Y + dH. \end{aligned}$$

It is immediate that u is a section of G_1 if and only if $\det(u)$ is invertible and the following relations hold:

$$\begin{aligned} (1) \quad a(d-1) &= ec, & (1') \quad a'(d-1) &= e'c', \\ (2) \quad b(d+1) &= ec', & (2') \quad b'(d+1) &= e'c, \\ (3) \quad e &= 2t(bc - ac'), & (3') \quad e' &= 2t(b'c' - a'c), \\ (4) \quad 2tc &= eb' - ae', & (4') \quad 2tc' &= be' - ea', \\ (5) \quad 2t(aa' - bb') &= 2td. \end{aligned}$$

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Lemma 5.3.— Relations (1), (1'), (2), and (2') imply

$$\det(u) = aa'(2-d) + bb'(d+2), \quad aa' - bb' = d \det(u).$$

Indeed, the first assertion follows at once by substituting relations (1), (1'), (2), and (2') in the expansion of $\det(u)$:

$$\begin{aligned} \det(u) &= aa'd + be'c + b'c'e - a'ec - ae'c' - bb'd \\ &= aa'd + bb'(d+1) + bb'(d+1) - aa'(d-1) - bb'd - aa'(d-1) \\ &= aa'(d-d+1-d+1) + bb'(d+1+d+1-d) \\ &= aa'(2-d) + bb'(d+2). \end{aligned}$$

³⁸Editor's note: the equality $[u(X), u(Y)] = 2tu(H)$ gives relations (4), (4'), and (5); the equality $[u(H), u(X)] = u(X)$ gives (1)–(3); and the equality $[u(H), u(Y)] = -u(Y)$ gives (1')–(3').

Multiplying this relation by d , we obtain

$$d \det(u) = aa'(2d - d^2) + bb'(d^2 + 2d).$$

On the other hand, the relation $(1) \times (1') = (2) \times (2')$ gives at once

$$aa'(d - 1)^2 = bb'(d + 1)^2.$$

Combining the last two relations immediately yields the second formula of Lemma 5.3.

5.4.— Now consider

$$G = G_1 \cap \mathrm{SL}(\mathfrak{g}).$$

This is the closed subgroup of G_1 defined by the equation $\det(u) = 1$; hence it is an affine group over S .

Proposition 5.5.— *The group G is smooth over S .*

To prove the proposition, we shall need the following lemmas.

Lemma 5.6.— *Let U be the open subset of*

$$\mathrm{End}_A(\mathfrak{g}) \simeq \mathbb{W}(M_3(\mathcal{O}_S))$$

defined by the condition that ad be invertible; that is, the principal open subset $\mathrm{End}_A(\mathfrak{g})_f$, where $f(u) = ad$. Then $U \cap G$ is the closed subscheme of U defined by the six equations

$$(1), \quad (2), \quad (2'), \quad (3), \quad (3'), \quad (\mathrm{D}): aa' - bb' = d.$$

It is clear that these six relations are satisfied by every point of G , by Lemma 5.3, and therefore by every point of $U \cap G$. Conversely, suppose that $u \in U(S')$, for some $S' \rightarrow S$, satisfies the six conditions in the statement. We must show that $\det(u) = 1$, and that u also satisfies $(1')$, (4) , $(4')$, and (5) .

First, (D) implies (5) . Relations (2) and $(2')$ give

$$bb'(d + 1) = bce' = b'c'e.$$

On the other hand, from (3) and $(3')$, writing $2t(bc - ac')(b'c' - a'c)$ in two ways, we obtain

$$(bc - ac')e' = (b'c' - a'c)e.$$

Together with the preceding relation, this gives

$$ac'e' = a'ce.$$

By (1) , however,

$$a'ce = a'a(d - 1).$$

Thus

$$a(a'(d - 1) - e'c') = 0,$$

and this implies $(1')$, since a is invertible.

Consequently, (1) , (2) , $(2')$, and $(1')$ hold. Lemma 5.3 and (D) now give

$$d(\det(u) - 1) = 0.$$

Since d is invertible, it follows that $\det(u) = 1$.

It remains to prove (4) and (4'). We prove (4'); the other calculation follows by symmetry. From (3), (3'), and (D), we have

$$a'e + be' = -2t(aa' - bb')c' = -2tdc'.$$

Combining (1') and (2), we also have

$$a'e + be' = -d(be' - ea'),$$

³⁹ which proves (4'), since d is invertible.

Lemma 5.7.— *The group G is smooth over S along the identity section.*

Proof. By 5.6 and SGA 1, II 4.10, it is enough to prove that the differentials of the functions

$$\begin{aligned} a(d-1) - ec, & \quad b(d+1) - ec', & \quad b'(d+1) - e'c, \\ e - 2t(bc - ac'), & \quad e' - 2t(b'c' - a'c), & \quad aa' - bb' - d \end{aligned}$$

at the points of the identity section of G are linearly independent. If the differential of a lower-case letter is denoted by the corresponding capital letter, these differentials are

$$D, \quad 2B, \quad 2B', \quad E + 2tC', \quad E' + 2tC, \quad A + A' - D,$$

⁴⁰ and they are indeed linearly independent modulo every $(t - \lambda)$, $\lambda \in k$. ⁴¹

Lemma 5.8.— *For $s \in S$, $s \neq s_0$, the fiber G_s is connected and semisimple.*

Proof. Indeed, since $s \neq s_0$, $\mathfrak{g}(s)$ is isomorphic to the Lie algebra of $\mathrm{PGL}_{2,\kappa(s)}$, and, on the other hand, $G_s = (G_1)_s$. The automorphism group of the Lie algebra of PGL_2 over a field of characteristic 0 is known to be PGL_2 itself, which is connected and semisimple. ⁴²

Lemma 5.9.— *The fiber G_{s_0} is solvable and has two connected components, of the following form:*

$$G_{s_0}^0 = \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & a^{-1} & 0 \\ c & c' & 1 \end{pmatrix} \right\}, \quad G_{s_0}^- = \left\{ \begin{pmatrix} 0 & b & 0 \\ b^{-1} & 0 & 0 \\ c & c' & -1 \end{pmatrix} \right\}.$$

Proof. We have $e = e' = 0$, since $t = 0$ at s_0 . The equations (1), (1'), ..., (5), and (D) can then be solved immediately.

Lemma 5.10.— *The matrix*

$$w = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

is a section of G over S , such that $w(s_0) \in G_{s_0}^-$.

Let us now prove Proposition 5.5. Denote by G^0 the union of the identity components of the fibers of G , that is, the complement of $G_{s_0}^-$. Since G is smooth over S along the identity section by Lemma 5.7, G^0 is an open subgroup of G , smooth over S , by Exposé VI_B, 3.10. By translation, G is also plainly smooth at the points of $w(S)$; hence G is smooth over S . ⁴³

³⁹Editor's note: the sign has been corrected.

⁴⁰Editor's note: $A + A' + B$ has been corrected to $A + A' - D$.

⁴¹Editor's note: " $(t - a)$, $a \in A$ " has been corrected to " $(t - \lambda)$, $\lambda \in k$ ".

⁴²Editor's note: the preceding sentence has been slightly modified.

⁴³Editor's note: the preceding sentence has been modified, and the reference to Exposé VI_B, 3.10 has been added.

5.11.— Consider the morphism

$$\mathbb{G}_{m,S} \longrightarrow G^0, \quad z \longmapsto \begin{pmatrix} z & 0 & 0 \\ 0 & z^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is a monomorphism defining a torus T of G^0 . We claim that

$$T = \text{Centr}_G(T) = \text{Centr}_{G^0}(T).$$

It is enough to verify the first equality. Since the two sides are smooth S -subgroups of G , it is enough to check that they have the same geometric points. In the fibers over points $s \neq s_0$, this follows because $\text{PGL}_{2,\kappa(s)}$ is reductive and T_s is a maximal torus in it for dimensional reasons (cf. 1.11). In the fiber over s_0 , the verification is immediate. In particular, T is a maximal torus of both G and G^0 .

5.12.— The section w of G defined in Lemma 5.10 normalizes T . It follows immediately (cf. 2.4) that the Weyl group of G is isomorphic to $(\mathbb{Z}/2\mathbb{Z})_S$, and is therefore finite over S :

$$W_G(T) = \text{Norm}_G(T)/T = (\mathbb{Z}/2\mathbb{Z})_S.$$

By contrast, $W_{G^0}(T)$ is not finite over S : it “lacks one point” above s_0 .

5.13.— The open immersion $G^0 \rightarrow G$ is not a closed immersion, since G^0 is dense in G ; it is nevertheless an affine morphism (and hence G^0 is affine over S). Indeed, $G^0_{s_0}$ is closed in G_{s_0} , which is closed in G , so its complement U in G is open. The open subsets G^0 and U cover G , and it is enough to verify that the immersions

$$G^0 \longrightarrow G^0 \quad \text{and} \quad G^0 \cap U \longrightarrow U$$

are affine. This is trivial for the first; for the second, observe that $U \cap G^0$ is defined in U by the condition $b \neq 0$.

We have thus constructed a smooth affine S -group G^0 , with connected fibers, possessing a self-centralizing maximal torus T whose Weyl group $W_{G^0}(T)$ is not finite (compare Theorem 2.5).

6. Local existence of maximal tori. The Weyl group

In the proof of Theorem 2.5 we used a result from Exposé XI on the local existence, for the étale topology, of maximal tori. The proof in Exposé XI uses a deep representability result (XI 4.1). In the particular case under consideration, one can give another, far more elementary proof, based on the ideas of Exposé XII, no. 7.

Proposition 6.1.— *Let S be a scheme, let G be a smooth affine S -group scheme with connected fibers over S , and let $s_0 \in S$ be a point such that the maximal tori of the geometric fiber $G_{\bar{s}_0}$ are self-centralizing. There exist an étale morphism $S' \rightarrow S$ whose image contains s_0 and a split maximal torus T of $G_{S'}$.*

Proof. First, we may suppose that S is affine.⁴⁴ Since G is of finite presentation over S , we may successively suppose that S is noetherian, local, and then henselian with separably closed residue field (cf. EGA IV, 8.12, §§8.8 and 18.8). Thus let

$$S = \text{Spec}(A), \quad k = \kappa(s_0),$$

⁴⁴Editor’s note: the preceding sentence and the reference to EGA IV below have been added.

where A is henselian and k is separably closed. Choose a maximal torus T_0 of $G_0 = G_k$. Such a torus exists, for example because the scheme of maximal tori of G_0 is smooth over k (Exposé XII, 7.1(c)). Since k is separably closed, T_0 is split (cf. Exposé X, 1.4), and is therefore given by a group monomorphism

$$f_0: \mathbb{G}_{m,k}^r \longrightarrow G_0.$$

Let $m > 1$ be an integer prime to the characteristic of k . By Exposé VIII, 6.7, for every $h > 0$, $\text{Centr}_{G_0}(m^h T_0)$ is represented by a closed subscheme of G_0 . The subgroup schemes $m^h T_0$ are schematically dense in T_0 (cf. Exposé IX, 4.10), and G_0 is noetherian. Consequently there is an h such that

$$\text{Centr}_{G_0}(m^h T_0) = \text{Centr}_{G_0}(T_0) = T_0.$$

Put $n = m^h$. Since n is invertible on S , ${}_n\mathbb{G}_{m,S}$ is isomorphic to $(\mathbb{Z}/n\mathbb{Z})_S$, so f_0 defines a group monomorphism

$$u_0: (\mathbb{Z}/n\mathbb{Z})_k^r \longrightarrow G_0$$

such that $\text{Centr}_{G_0}(u_0) = T_0$.

Now the S -functor

$$P = \text{Hom}_{S\text{-gr}}((\mathbb{Z}/n\mathbb{Z})_S^r, G)$$

is represented by an S -scheme of finite type: it is a closed subscheme of r copies of

$${}_nG = \text{Hom}_{S\text{-gr}}((\mathbb{Z}/n\mathbb{Z})_S, G) = \text{Ker}(G \xrightarrow{n} G).$$

But P is smooth over S (Exposé IX, 3.6). Therefore $u_0 \in P(k)$ lifts to a section $u \in P(S)$, by Hensel's lemma (Exposé XI, 1.11):

$$u: (\mathbb{Z}/n\mathbb{Z})_S^r \longrightarrow G.$$

Consider $H = \text{Centr}_G(u)$. It is a closed subgroup scheme of G , by Exposé VIII, 6.5(e), and its special fiber is $H_0 = T_0$. Moreover, H is smooth over S . Indeed, let $S' = \text{Spec}(A')$ be an affine S -scheme, let

$$u': ({}_n\mathbb{G}_{m,S'})^r \longrightarrow G_{S'}$$

be obtained from u by base change, put $S'_J = \text{Spec}(A'/J)$, where J is a square-zero ideal, and let $x \in H(S'_J)$. Since G is smooth, x lifts to an element $g \in G(S')$. Then

$$v = \text{int}(g)(u')$$

satisfies $v_J = u'_J$. By Exposé IX, 3.2, there is therefore an element $g' \in G(S')$ such that $g'_J = e$ and $\text{int}(g')(v) = u'$. It follows that $h = g'g$ belongs to $H(S')$ and satisfies $h_J = x$.
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Let H^0 be the identity component of H . It is a subgroup scheme of G , smooth and with connected fibers, whose special fiber is a torus. By Exposé X, 8.1, it is a torus, necessarily split by Exposé X, 4.6. Put $T = H^0$, and let

$$C = \text{Centr}_G(T).$$

This is a closed subgroup of G (Exposé VIII, 6.5(e)) and is smooth (Exposé XI, 2.4). Consider C^0 (in fact $C^0 = C$, but we do not need this). Then $C^0 \supset T$, and both are

⁴⁵Editor's note: the reference to Exposé VIII, 6.5(e) was added above, and the preceding smoothness argument has been expanded.

smooth groups with connected fibers. They coincide at s_0 , hence in a neighborhood of s_0 . After shrinking S , we may therefore suppose that $C^0 = T$; then, a fortiori, T is maximal.

Remark 6.2.— *The proof shows in particular that the reductive rank of $G_{\bar{s}}$ is constant in a neighborhood of $s = s_0$.*

Proposition 6.3.— *Let S be a scheme, let G be an S -group scheme that is smooth and of finite presentation over S , and let Q be a subtorus of G .*

- (i) $\text{Centr}_G(Q)$ and $\text{Norm}_G(Q)$ are represented by closed subgroup schemes, smooth (and hence of finite presentation) over S .
- (ii) $\text{Centr}_G(Q)$ is an open and closed subscheme of $\text{Norm}_G(Q)$. The quotient

$$W_G(Q) = \text{Norm}_G(Q) / \text{Centr}_G(Q)$$

is represented by an open subgroup scheme of $\text{Aut}_{S\text{-gr}}(Q)$; it is therefore a quasi-finite, étale, separated S -group scheme.

- (iii) For every $s \in S$, put

$$w(s) = \left| \text{Norm}_{G(\bar{s})}(Q(\bar{s})) / \text{Centr}_{G(\bar{s})}(Q(\bar{s})) \right|.$$

Then $s \mapsto w(s)$ is lower semicontinuous, and is constant in a neighborhood of s if and only if $W_G(Q)$ is finite over S in a neighborhood of s .

Proof. By Exposé XI, 6.11, $\text{Centr}_G(Q)$ and $\text{Norm}_G(Q)$ are represented by closed subschemes of G of finite presentation. They are smooth by Exposé XI, 2.4 and 2.4 bis, which proves (i). Assertions (ii) and (iii) are then proved as in Exposé XI, 5.9 and 5.10; those proofs in fact use only (i), and not the deeper theorems of Exposé XI, 4.1 and 4.2.

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